2022 VJC Prelim H2 P3 Suggested solution

1(a) At maximum height,

Loss in KE of mass = gain in GPE of mass + gain in EPE in cord [CONCEPT]

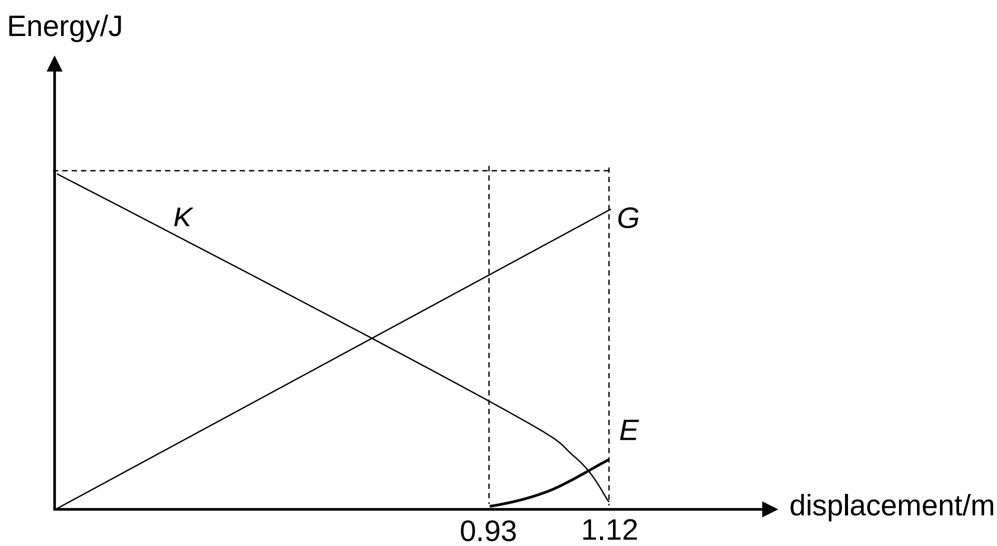
$$\frac{1}{2}mv^2 - 0 = mgh + \frac{1}{2}kx^2$$

$$\frac{1}{2}(150\times10^{-3})(5.7^2) - 0 = (150\times10^{-3})(9.81)(1.12) + \frac{1}{2}(45)x^2$$

$x = 0.19$ m

(b) Length of unstretched cord = 1.12 – 0.19 = 0.93 m

(c)



(G should end lower so that G + E = initial K at 1.12 m)

(d) Since the tension acting on the ball is always towards the center and perpendicular to the displacement of the ball [APPLICATION], there is no work done by the tension. [CONCLUSION]
[CONCEPT: WD = force x displacement in the direction of the force]

2(a) When the sound wave hits the wall, it will be reflected in the opposite direction and